Scatter Plots and Predictions

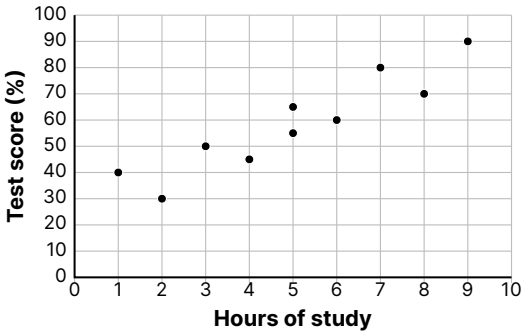
Name _____ **Date** _____ Calculator allowed **Version 1**

KEY WORDS scatter plot · relationship · positive · negative · strong · weak · line of best fit · prediction · interpolation · extrapolation · outlier

A Reading a scatter plot ●○○ — ANSWER ON THE SHEET

Remember: Each point shows two measurements for one person or thing. Read across to one axis and down to the other. Always check what the axes measure and their units.

1. The scatter plot shows the hours of study and the test scores of some students.



a) How many students are shown? _____

b) What is the highest test score, and how many hours did that student study? _____

c) What score did the student who studied for 2 hours get? _____

d) How many students scored more than 60%? _____

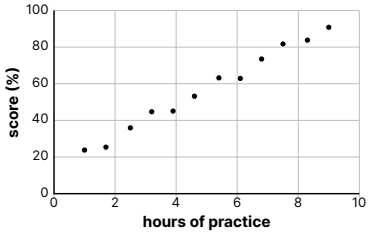
e) How many students studied for less than 4 hours? _____

B Writing the relationship sentence ●●○ — WORK IN YOUR BOOK

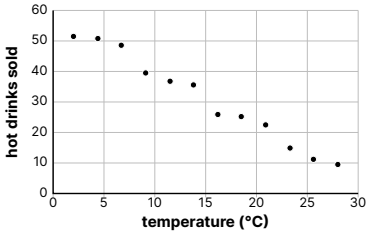
Remember: Say three things: how strong (strong or weak), which way (positive or negative), and between what and what, using the names on the axes: "There is a strong positive relationship between hours of practice and score."

2. For each scatter plot, write the full sentence that describes the relationship, or write that there is no relationship.

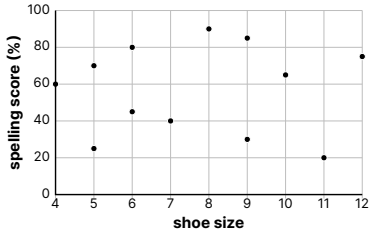
a)



b)

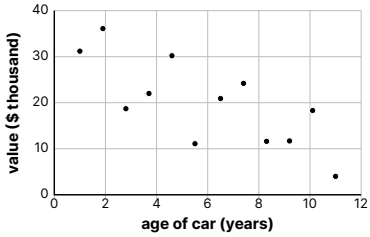


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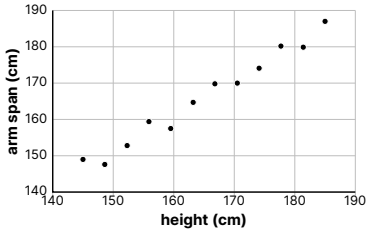


3. Again, write the full sentence for each. Two of these are weak relationships: say so.

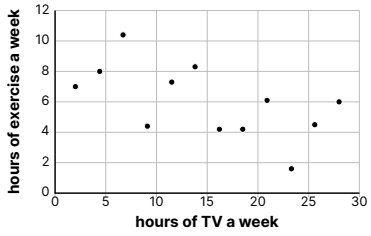
a)



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c)



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C What the relationship means ●●○ WORK IN YOUR BOOK

Remember: Turn the sentence into plain words about the situation: "As the hours of practice go up, the score tends to go up." Say "tends to", because a scatter plot shows a pattern, not a rule.

4. For each relationship, write what it means in the situation, starting with "As ...".

a) A strong positive relationship between hours of practice and music exam mark.

b) A strong negative relationship between the temperature and the number of hot drinks sold.

c) A weak negative relationship between the age of a car and its value.

d) No relationship between shoe size and spelling score.

D The line of best fit ●●● WORK IN YOUR BOOK

Remember: A line of best fit follows the trend of the points, with about as many points above it as below. Put a value into its equation to make a prediction.

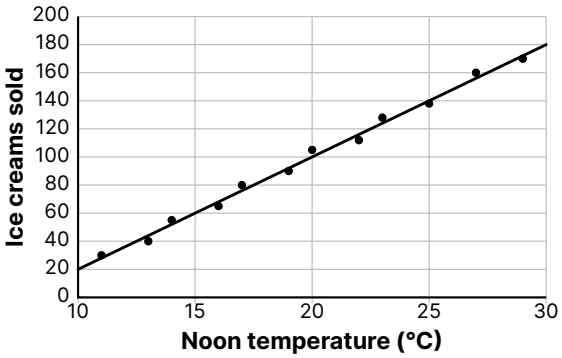
5. A shop records the noon temperature and the number of ice creams it sells on twelve days. The line of best fit is $n = 8t - 60$.

a) Write the sentence that describes the relationship.

b) Predict the number of ice creams sold when the temperature is 18°C.

c) Predict the number of ice creams sold when the temperature is 24°C.

d) What temperature gives a predicted number of ice creams sold of 100?



E Interpolation and extrapolation ●●● WORK IN YOUR BOOK

Remember: A prediction inside the range of the data is interpolation. A prediction outside it is extrapolation, which is less reliable because we have not seen what happens there.

6. Use the data in the question above. Is a prediction for each of these values interpolation or extrapolation?

a) a temperature of 20°C

b) a temperature of 35°C

c) a temperature of 12°C

d) a temperature of 8°C

7. The line is used to predict the number of ice creams sold when the temperature is 40°C.

a) What does the line predict?

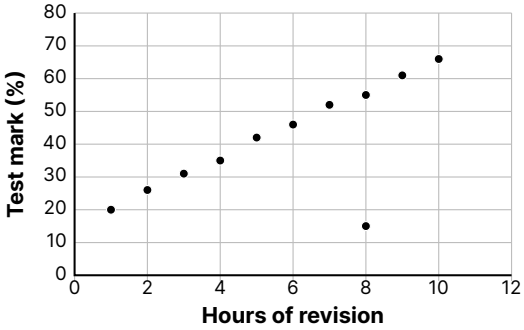
b) Explain why this prediction may not be reliable.

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F Outliers and cause ●●● WORK IN YOUR BOOK

Remember: An outlier is a point far from the pattern of the others. A relationship on a scatter plot does not prove that one thing causes the other: something else may be behind both.

8. The scatter plot shows the hours a group of students spent revising and their test marks.



a) One point is an outlier. Give its coordinates.

b) Suggest a reason for the outlier.

c) Ignoring the outlier, write the sentence that describes the relationship.

d) A student says: "This proves that revising for more hours causes a higher mark." Is that a fair conclusion? Explain.

9. Over ten summers, ice cream sales and the number of people treated for sunburn both went up and down together, giving a strong positive relationship. Does eating ice cream cause sunburn? What is really going on?

Extension FOR EARLY FINISHERS

10. In the equation of the line of best fit for the ice cream data, what does the number 8 tell you about the number of ice creams sold?

11. The line predicts −60 ice creams at 0°C. What is wrong with this, and why does it happen?

12. Two scatter plots both show a positive relationship. On one the points lie close to a line; on the other they are widely scattered. Which gives the more reliable predictions, and why?

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Scatter Plots and Predictions

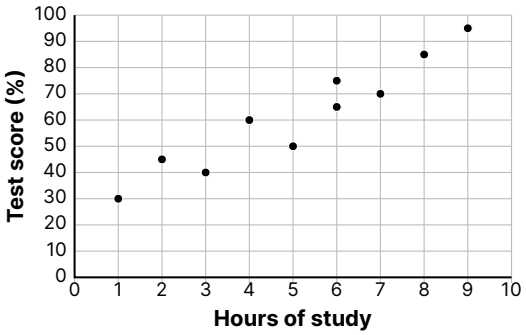
Name _____ **Date** _____ Calculator allowed **Version 2**

KEY WORDS scatter plot · relationship · positive · negative · strong · weak · line of best fit · prediction · interpolation · extrapolation · outlier

A Reading a scatter plot ●○○ — ANSWER ON THE SHEET

Remember: Each point shows two measurements for one person or thing. Read across to one axis and down to the other. Always check what the axes measure and their units.

1. The scatter plot shows the hours of study and the test scores of some students.



a) How many students are shown? _____

b) What is the highest test score, and how many hours did that student study? _____

c) What score did the student who studied for 3 hours get? _____

d) How many students scored more than 70%? _____

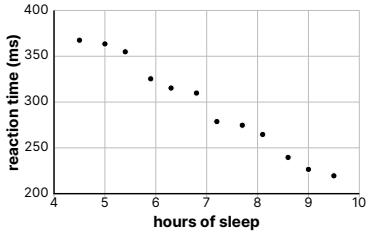
e) How many students studied for less than 5 hours? _____

B Writing the relationship sentence ●●○ — WORK IN YOUR BOOK

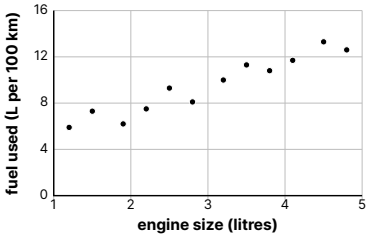
Remember: Say three things: how strong (strong or weak), which way (positive or negative), and between what and what, using the names on the axes: "There is a strong positive relationship between hours of practice and score."

2. For each scatter plot, write the full sentence that describes the relationship, or write that there is no relationship.

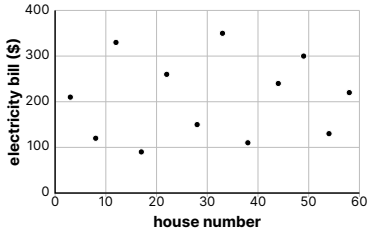
a)



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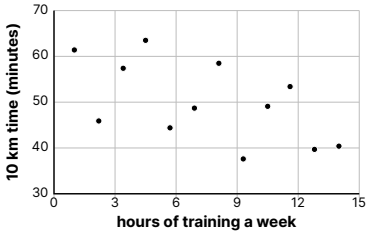


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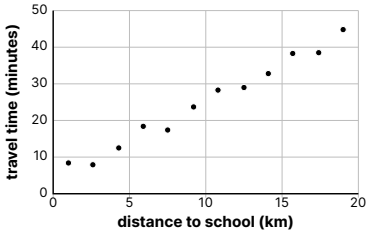


3. Again, write the full sentence for each. Two of these are weak relationships: say so.

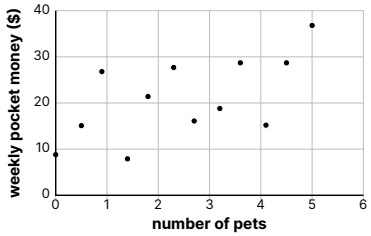
a)



b)



c)



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C What the relationship means ●●○ WORK IN YOUR BOOK

Remember: Turn the sentence into plain words about the situation: "As the hours of practice go up, the score tends to go up." Say "tends to", because a scatter plot shows a pattern, not a rule.

4. For each relationship, write what it means in the situation, starting with "As ...".

a) A strong negative relationship between hours of sleep and reaction time.

b) A strong positive relationship between distance to school and travel time.

c) A weak positive relationship between the number of pets and weekly pocket money.

d) No relationship between house number and electricity bill.

D The line of best fit ●●● WORK IN YOUR BOOK

Remember: A line of best fit follows the trend of the points, with about as many points above it as below. Put a value into its equation to make a prediction.

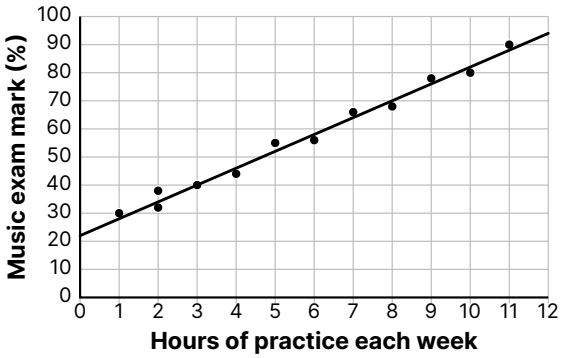
5. A music teacher records the hours her students practise each week and their exam marks. The line of best fit is $m = 6h + 22$.

a) Write the sentence that describes the relationship.

b) Predict the exam mark when the practice time is 3 hours.

c) Predict the exam mark when the practice time is 8 hours.

d) What practice time gives a predicted exam mark of 82%?



E Interpolation and extrapolation ●●● WORK IN YOUR BOOK

Remember: A prediction inside the range of the data is interpolation. A prediction outside it is extrapolation, which is less reliable because we have not seen what happens there.

6. Use the data in the question above. Is a prediction for each of these values interpolation or extrapolation?

a) a practice time of 5 hours

b) a practice time of 14 hours

c) a practice time of 10 hours

d) a practice time of 0.5 hours

7. The line is used to predict the exam mark when the practice time is 15 hours.

a) What does the line predict?

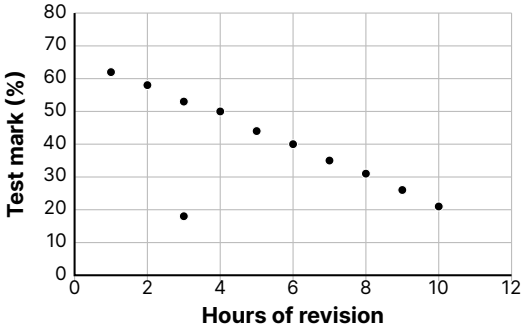
b) Explain why this prediction may not be reliable.

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b) Suggest a reason for the outlier.

c) Ignoring the outlier, write the sentence that describes the relationship.

d) A student says: "This proves that revising for more hours causes a higher mark." Is that a fair conclusion? Explain.

9. Over ten summers, ice cream sales and the number of people treated for sunburn both went up and down together, giving a strong positive relationship. Does eating ice cream cause sunburn? What is really going on?

★ Extension FOR EARLY FINISHERS

10. In the equation of the line of best fit for the music exam data, what does the number 6 tell you about the exam mark?

11. The line predicts a mark of 22% for a student who does no practice. Is this prediction interpolation or extrapolation? How much would you trust it?

12. Two scatter plots both show a positive relationship. On one the points lie close to a line; on the other they are widely scattered. Which gives the more reliable predictions, and why?

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